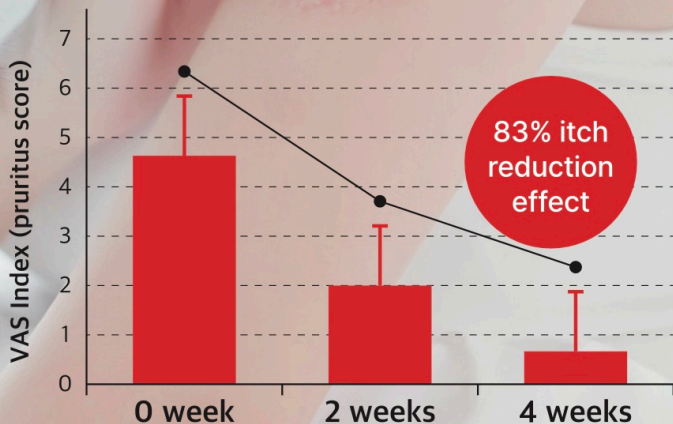


# Next powered Solution **Science for Hair & Skin**

RA:CLIA

# BARRICURE<sup>®</sup>

is for improving "itchy skin"



\* Clinical trial is the result of university hospital (dermatology) in Korea, Seoul

# STEMIR 22

Hair Booster

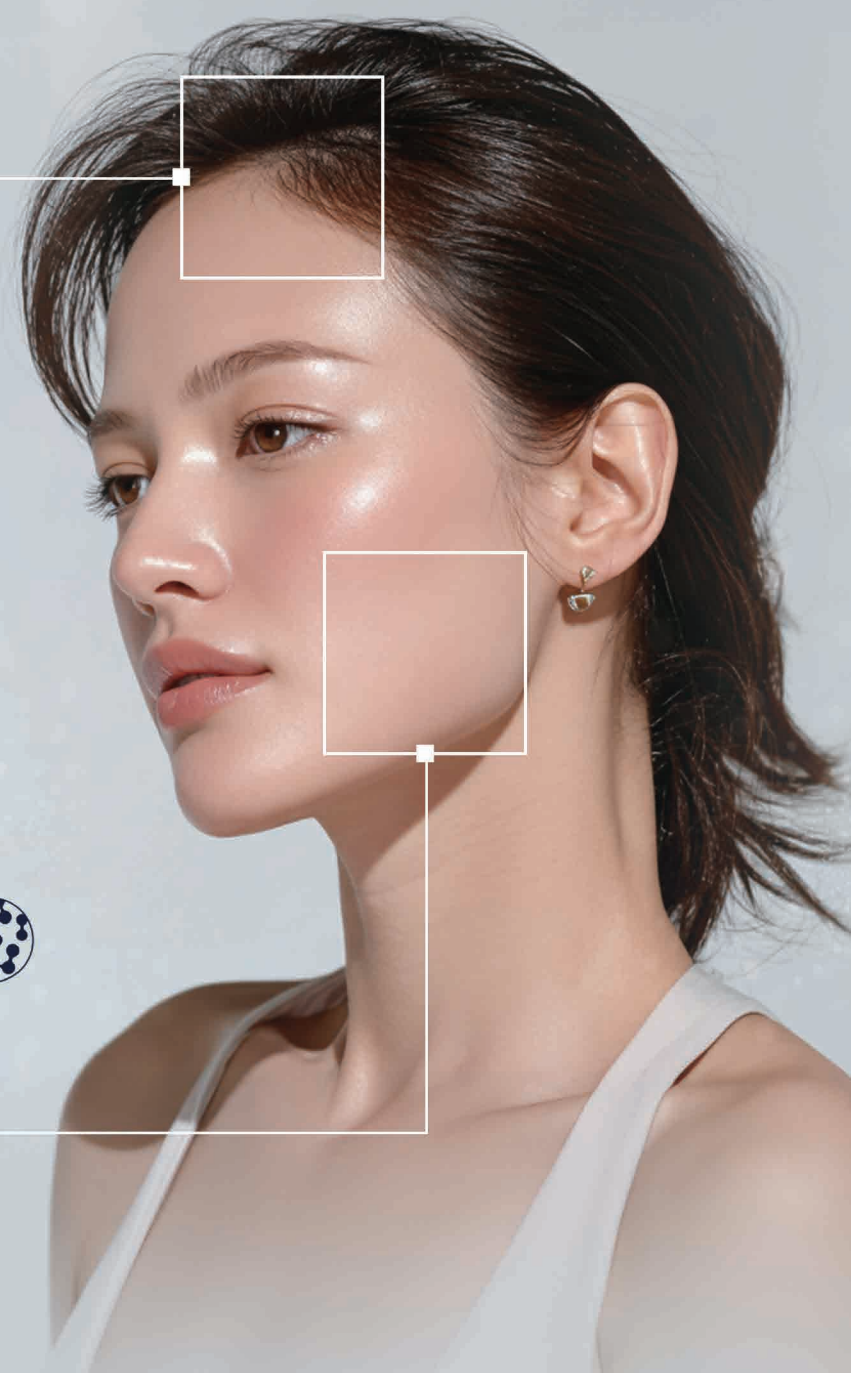
22-Based  
miRNA



# STEMIDE 31

Skin Booster

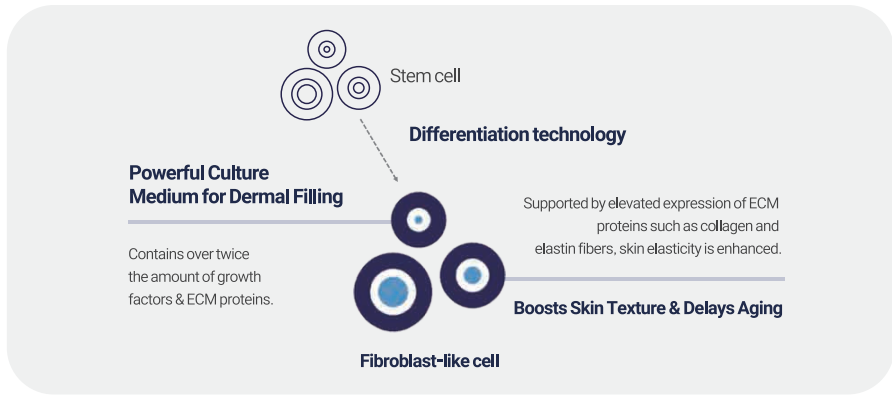
4 Types of Growth  
Factor Peptides



# Special technology 1. Nexotide™

## 1. What is Nexotide™

Nexotide™ is a skin rejuvenation platform that uses biomimic technology to reproduce the benefits of stem cell culture medium. By differentiating stem cells into fibroblast-like cells, we produce a culture medium rich in key growth factors—EGF, TGF- $\beta$ , Thymopoietin, and Collagen Fragments—at over twice the amount found in conventional fibroblast medium. **Nexotide™ offers the same effects through stable, easily absorbed biomimetic peptides for enhanced and faster skin improvement.**



## 2. The Story of Nexotide's Birth

1

### Stem Cell Development

hADSD isolation induces fibroblast-like cell

2

### Discovery of Efficacy of the Conditioned Medium

3

### Profiling: Key Growth Factors

EGF, TGF- $\beta$ , Thymopoietin, Collagen Fragments

4

### Birth of Nexotide

Safe Biomimetic Culture Medium Developed

**Nexotide™ technology has been proven through patents and academic papers.**

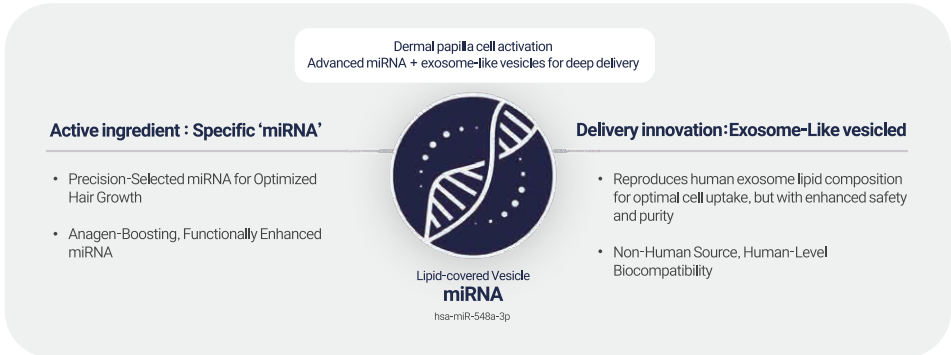




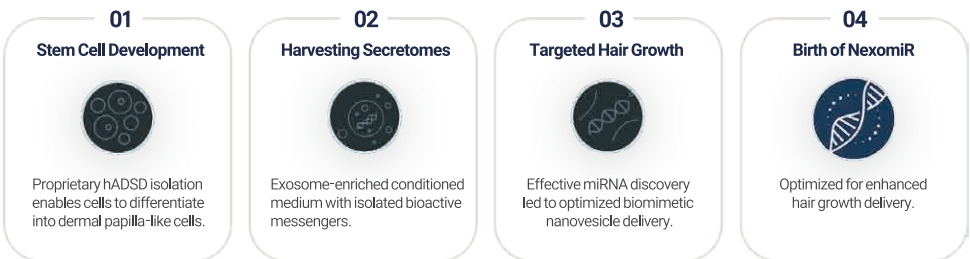
# Special technology 2. NexomiR™

## 1. What is NexomiR™

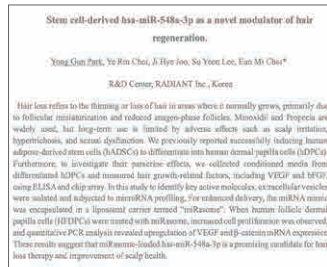
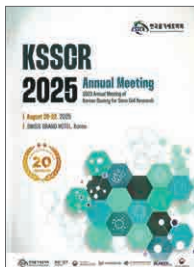
NexomiR™ is an advanced stem cell–derived miRNA technology for targeted hair regeneration.



## 2. The story of NexomiR's Birth



NexomiR™ technology has been proven through patents and academic papers



# Product 1. STEMIR22 Hair Booster

**STEMIR22 Hair Booster** is an advanced professional solution for hair growth and scalp health, featuring a specialized **22-base miRNA** sequence derived from stem cell exosomes.



1ml \* 5 syringe

Targets hair loss at the root level

Reactivates scalp vitality through miRNA technology

Promotes stronger, denser hair regeneration

## Key Active Components of STEMIR22

### 22-Based miRNA

- Dermal papilla cells Activation
- (Synthetic, non-human-derived)



### Niacinamide

- Dermal papilla cell growth promotion

### Red Ginseng

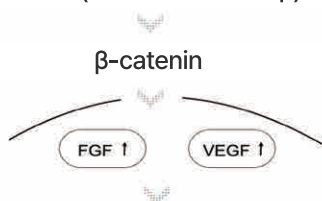
- Promotion of hair growth factor expression

### Hydrolyzed hyaluronic acid

- Scalp Moisturizing

## Mechanism of STEMIR22

miRNA (hsa-miR-548a-3p)



### Stimulates derma papilla cell growth

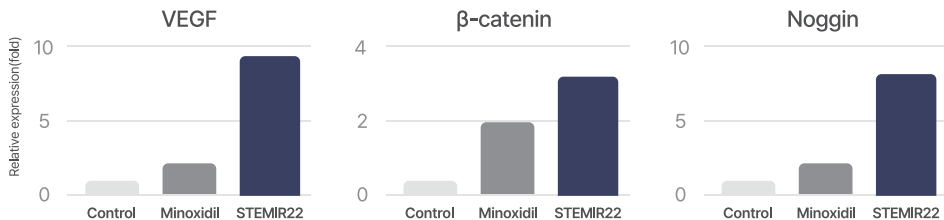
STEMIR22 promotes hair follicle growth by activating  $\beta$ -catenin signaling and inducing VEGF and FGF expression.

- 1 Biomimetic vesicles deliver miRNA deep into hair follicles.
- 2 Activates  $\beta$ -catenin, FGF, and VEGF pathways.
- 3 Stimulates hair follicle regeneration for stronger, healthier hair.

# STEMIR22 Efficacy 1. In vitro, Ex vivo

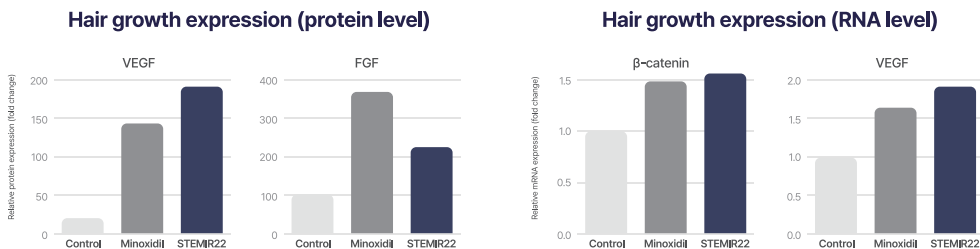
## STEMIR22 Promotes **Anagen Phase Hair Growth**

### In vitro : Activation of Derma papila cells



- Activates VEGF,  $\beta$ -catenin, and Noggin pathways to trigger the hair growth phase.

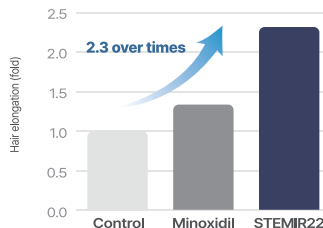
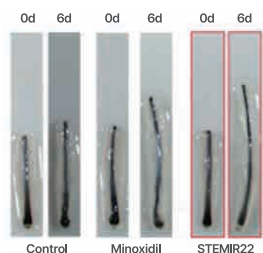
### In vitro : Hair Growth Factor Expression



- STEMIR22 upregulates  $\beta$ -catenin, VEGF, and FGF, promoting follicle regeneration

### Ex vivo : Hair Growth Stimulation

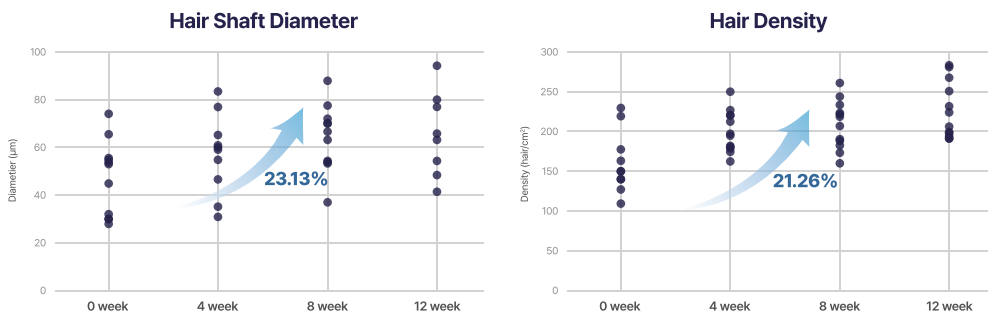
#### Anagen Induction (human organ)



- 2.3x greater hair growth in ex vivo human follicle model.

# STEMIR22 Efficacy 2. Clinical study

## 1. Clinically Proven Improvement in Hair Thickness and Density



- After 12 weeks of using STEMIR22, participants saw an average increase of 23.13% in hair thickness and 21.26% in density. Some individuals experienced maximum increases of up to 39.53% in thickness and 20.73% in density.

## 2. Visible Results After 12 Weeks of STEMIR22 Application

|                        | CASE 01                 |                         |                         | CASE 02                |                        |                         |
|------------------------|-------------------------|-------------------------|-------------------------|------------------------|------------------------|-------------------------|
| Measurement            | 0 weeks                 | 4 weeks                 | 12 weeks                | 0 weeks                | 4 weeks                | 12 weeks                |
| Top of the head        |                         |                         |                         |                        |                        |                         |
| Diameter of hair shaft | <br>33 µm (32.5 ± 18.5) | <br>34 µm (37.5 ± 12.5) | <br>41 µm (44.5 ± 20.5) | <br>29 µm (30.0 ± 9.0) | <br>31 µm (32.0 ± 6.0) | <br>34 µm (49.5 ± 26.5) |
| Hair density           | <br>176 hairs/cm²       | <br>193 hairs/cm²       | <br>228 hairs/cm²       | <br>209 hairs/cm²      | <br>242 hairs/cm²      | <br>271 hairs/cm²       |

- Representative clinical case: Sequential images show progressive enhancement in hair quality and density over a 12-week course of STEMIR22 treatment, as documented in a controlled clinical study.



# Product 2. STEMIDE31 Skin Booster

STEMIDE31 Skin Booster is an advanced injectable that improves skin elasticity and reduces wrinkles. It contains 4 biomimetic peptides for skin rejuvenation. These peptides total **31 amino acids** optimized for regeneration.



**3ml \* 5 vials**

## Volume Improvement

- Reduces fine lines
- Boosts skin elasticity

## Skin Enhancement

- Refines skin texture
- Deep hydration from within
- Radiant skin confidence

## Key Active Components of STEMIDE31

### 4 Types of Growth Factor Peptides

(EGF, TGF- $\beta$ , Thymopoietin, and Collagen Fragments)  
-Skin Rejuvenation

(Synthetic, non-human-derived)



### 1X10<sup>10</sup> Lactobacillus Exosomes

- Boosts collagen production

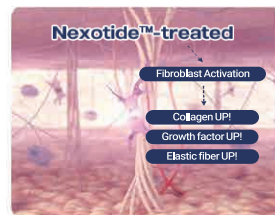
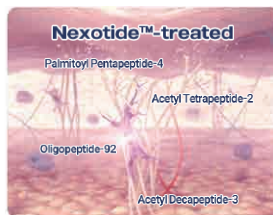
### 1% PDRN

- Regeneration

### Glutathione

- Whitening

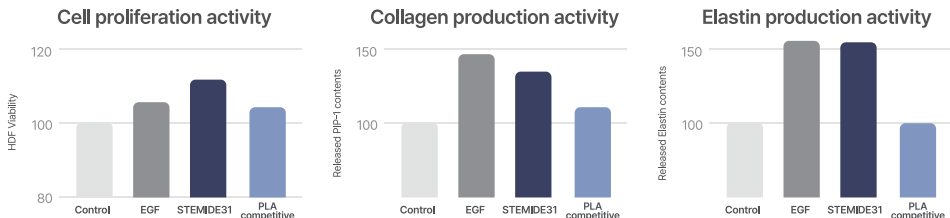
## Mechanism of STEMIDE31



- 1 STEMIDE31 activates fibroblasts.
- 2 Activated fibroblasts increase collagen, growth factors, and elastic fibers.
- 3 It promotes skin regeneration and enhances skin elasticity.

# STEMIDE31 Efficacy 1. In vitro

## In vitro : Enhanced Fibroblast Activity and Dermal Regeneration

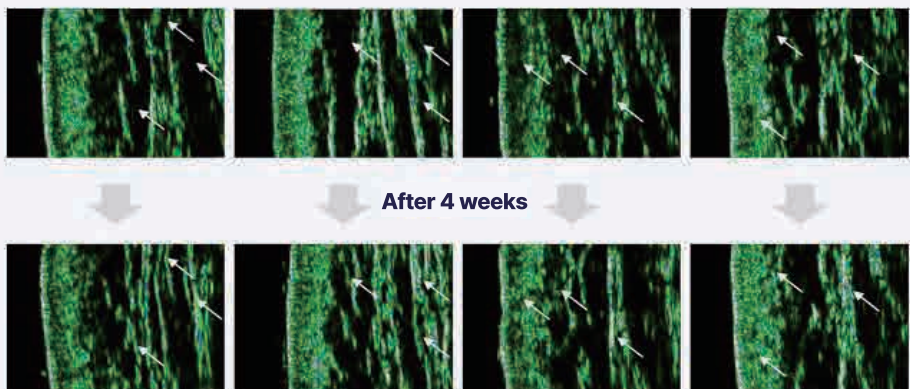


- Increased cell proliferation 115%, collagen synthesis 122%, elastin production 137%.
- Compared to competing PLA-containing products, superior fibroblast activity was observed.

► Strong efficacy in skin regeneration, elasticity improvement, and dermal revitalization.

## STEMIDE31 Efficacy 2. Clinical study

### 1. Measurement of skin density

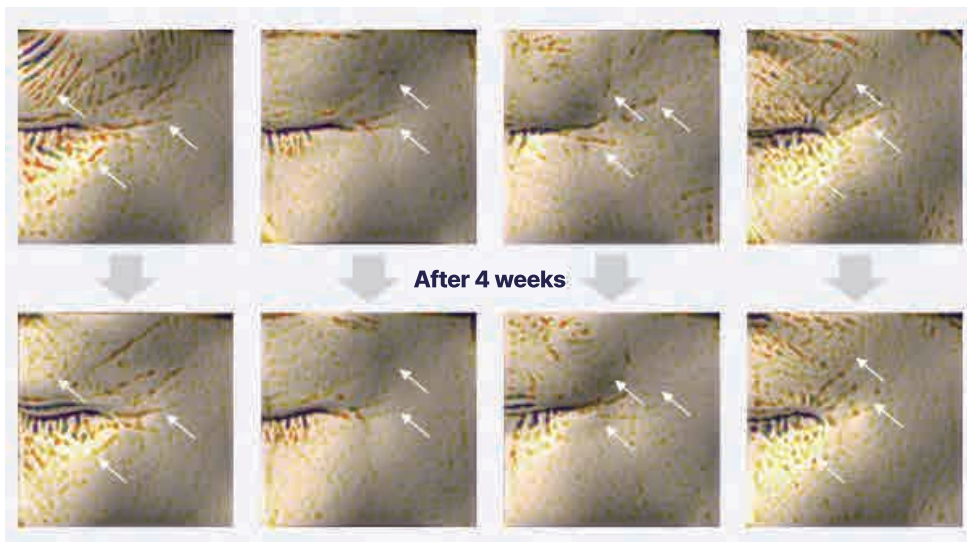


\*A wider, clearer, and more uniform green area indicates higher dermal density

► After 4 weeks of STEMIDE31 application, skin density **increased by 30.2%**

## 2. Measurement of wrinkles around the eyes

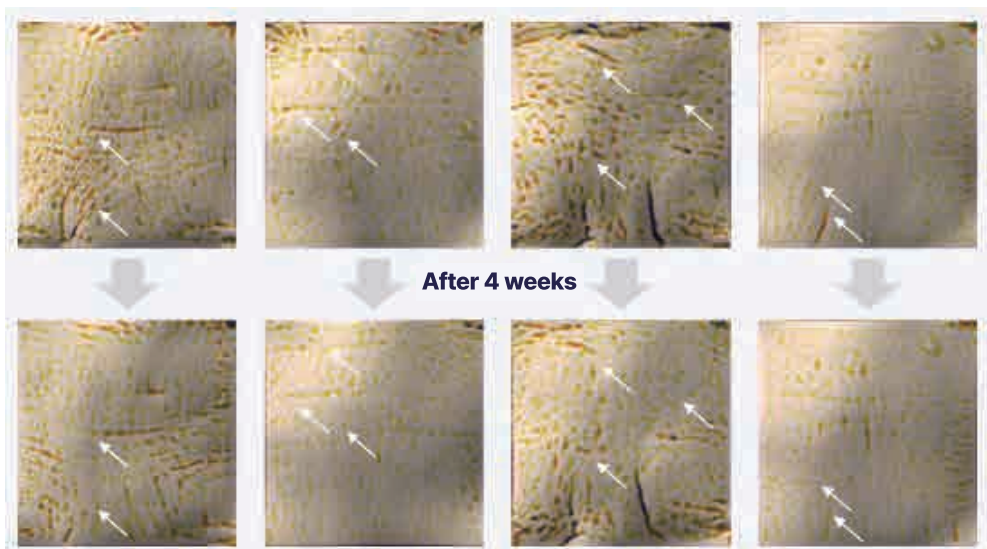
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► A 4-week clinical test showed a **28.2% reduction** in eye wrinkles after STEMIDE31 application.

## 3. Forehead wrinkle measurement

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► A 4-week clinical test showed a **32.1% improvement** in forehead wrinkles after STEMIDE31 application.

# HOW TO USE

## 1. STEMIR22 Hair Booster



- 1 syringe per treatment (1ml)
- Over 12 weeks, with sessions administered every 1 to 2 weeks

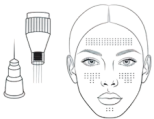
A professional applies the product to the scalp, followed by MTS, AIR JET, or other supplementary treatments to enhance absorption.

## 2. STEMIDE31 Skin Booster



### STEP 01

Take the solution out of the vial with a syringe, or load it into a cartridge when using an AIR JET device.



### STEP 02

Apply the product using various methods, including auto injection, AIR JET, MTS, or manual administration.

- For the initial 3 sessions, have the treatment every 1 week.
- After the initial phase, continue with treatments once a month to maintain your skin.

## Next-Powered Solution by RADIANT Inc

### Pioneering Innovation in Bio-materials 'RADIANT Inc'

#### Stem Cell Aesthetic Technology

Since 2015, we have patented stem cell technologies to create fibroblasts and dermal papilla cells from adipose-derived stem cells.

Through our advancements in stem cell technology, we have ultimately developed the **Next powered solutions—Nexotide™ and NexomiR™ technologies**



#### Global biotechnology company

Since 2002, RADIANT Inc. has been a leading company specializing in the development of advanced bio-materials.

#### Innovative Skin Solutions

10,000 efficacious skin ingredients have been developed using diverse natural resources, advancing cosmetic science.

#### AI-Powered Biotech

100,000 natural resources are integrated with AI to accelerate next-generation bio-material development.



# STEMIR22 Hair Booster

# STEMIDE31 Skin Booster

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